MA3236 NONLINEAR PROGRAMMING
Semester 1, AY 2026/2027
Tutorial 7

Review: Lecture 14 — convex KKT and SVM

Consider

$$\min_{\mathbf{x}\in\mathcal{D}} f(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathbf{h}(\mathbf{x}) \le \mathbf{0},$$

where $\mathcal{D}$ is open convex, $f, h_j \in C^1$ are convex, and g_i are affine. Write $\mathcal{C}$ for the feasible set.

Certifying a candidate. For a feasible $\mathbf{x}^*$, use $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T\mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T\mathbf{h}(\mathbf{x})$. Seek free equality multipliers $\boldsymbol{\lambda}^*$ and inequality multipliers $\boldsymbol{\nu}^*$ satisfying

$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0}, \qquad \boldsymbol{\nu}^* \ge \mathbf{0}, \qquad \nu_j^* h_j(\mathbf{x}^*) = 0 \quad (\forall j).$$

These are KKT, with feasibility already imposed. Convexity makes them sufficient for global optimality without a CQ. Conversely, LICQ at a minimizer gives KKT and unique multipliers. An active constraint may still have zero multiplier.

Separating existence from uniqueness.

- Existence from previous lectures: continuous f on nonempty compact $\mathcal{C}$ ensures attainment; strict convexity on $\mathcal{C}$ gives at most one minimizer, but not attainment.
- For convex C^2 data, KKT + SOSC also gives a unique global minimum. Strict convexity and SOSC do not imply each other in general; primal uniqueness need not give multiplier uniqueness.

Using the certificate for hard-margin SVM. For labeled data $(\mathbf{x}_i, y_i)$, $y_i \in \{-1, 1\}$, normalizing the margins to $\mathbf{w}^T\mathbf{x} + b = \pm 1$ gives width $2/\|\mathbf{w}\|$. Maximizing it gives

$$\min_{\mathbf{w},b} \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad h_i(\mathbf{w}, b) = 1 - y_i(\mathbf{w}^T\mathbf{x}_i + b) \le 0 \quad (i = 1, \dots, N).$$

This is a convex program. Differentiating its Lagrangian in $\mathbf{w}, b$ gives the certificate

$$\mathbf{w} = \sum_i \nu_i y_i \mathbf{x}_i, \qquad \sum_i \nu_i y_i = 0, \qquad h_i \le 0, \quad \nu_i \ge 0, \quad \nu_i h_i = 0.$$

The separator is $\mathbf{w}^T\mathbf{x} + b = 0$; $\nu_i > 0$ identifies a loaded support vector on a margin (the converse can fail). Both classes and strict separability give a unique $(\mathbf{w}^*, b^*)$, but not necessarily unique multipliers.

Allowing margin violations. If hard-margin constraints are infeasible, introduce slacks $\xi_i \ge 0$ and penalize them with $C > 0$:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \tfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad 1 - y_i(\mathbf{w}^T\mathbf{x}_i + b) \le \xi_i, \quad \xi_i \ge 0.$$

At fixed $(\mathbf{w}, b)$, the smallest slack is $\xi_i^{\min} = [1 - y_i(\mathbf{w}^T\mathbf{x}_i + b)]_+$. Thus $\xi_i^{\min} = 0$ means on/outside the correct margin; $0 < \xi_i^{\min} < 1$ means inside it; $\xi_i^{\min} = 1$ means on the separator; and $\xi_i^{\min} > 1$ means on the wrong side. For finite data, the soft-margin model is convex, feasible, and attains a minimum.

Review: Lecture 15 — QP and ALM

For $\min f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$, let $f, g_i \in C^2$ on open $\mathcal{D}$. Write

$$L = f + \boldsymbol{\lambda}^T \mathbf{g}, \qquad \mathbf{A} = [\nabla g_1, \ldots, \nabla g_m], \qquad \mathbf{H}_L = \nabla^2_{\mathbf{xx}} L.$$

Quadratic penalty method. The quadratic penalty (QP) method uses an effective multiplier ($\rho > 0$) to interpret its derivatives:

$$P_\rho = f + \tfrac{\rho}{2}\|\mathbf{g}\|^2, \qquad \widehat{\boldsymbol{\lambda}} = \rho \mathbf{g},$$
$$\nabla P_\rho = \nabla f + \mathbf{A}\widehat{\boldsymbol{\lambda}}, \quad \nabla^2 P_\rho = \mathbf{H}_L(\mathbf{x}, \widehat{\boldsymbol{\lambda}}) + \rho \mathbf{A}\mathbf{A}^T.$$

- Inner stationarity supplies KKT stationarity, but not feasibility. At a LICQ KKT pair with $\boldsymbol{\lambda}^* \neq \mathbf{0}$, no finite ρ gives $\nabla P_\rho(\mathbf{x}^*) = \mathbf{0}$. The traditional rule increases $\rho^{(k+1)} = \gamma \rho^{(k)}$ with fixed $\gamma > 1$.
- Large normal curvature with bounded tangent curvature can cause ill-conditioning.

Augmented Lagrangian method. The augmented Lagrangian method (ALM) adds a linear correction: with fixed $\rho > 0$, minimize $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho)$, then update:

$$L_A = f + \boldsymbol{\lambda}^T \mathbf{g} + \tfrac{\rho}{2}\|\mathbf{g}\|^2, \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)}).$$

Substituting the update gives

$$\nabla_{\mathbf{x}} L_A(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho) = \nabla_{\mathbf{x}} L(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)}).$$

Inner stationarity uses the updated multiplier, so feasibility can improve at fixed ρ.

Separate inner and outer tests. Warm-start SD or Newton with line search on $\Psi^{(k)} = P_{\rho^{(k)}}$ or $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho)$, until

$$\|\nabla \Psi^{(k)}(\mathbf{x}^{(k+1)})\|_\infty \leq \varepsilon_{\text{in}}^{(k)} < \varepsilon_{\text{out}}, \qquad \varepsilon_{\text{in}}^{(k)} > 0.$$

With the QP estimate or updated ALM multiplier, require both KKT residuals:

$$r_{\text{KKT}} = \max\{\|\nabla f + \mathbf{A}\boldsymbol{\lambda}\|_\infty, \|\mathbf{g}\|_\infty\} \leq \varepsilon_{\text{out}}.$$

A small inner gradient alone cannot certify feasibility. A fixed inner tolerance does not in general guarantee exact asymptotic convergence.

Match guarantees to assumptions.

- For QP with $\rho^{(k)} \to \infty$ and a nonempty feasible set, exact global inner minima give globally optimal cluster points in $\mathcal{D}$. For stationary solves, bounded $\widehat{\boldsymbol{\lambda}}$ forces $\mathbf{g} = \widehat{\boldsymbol{\lambda}}/\rho \to \mathbf{0}$; vanishing inner gradients and a convergent primal–multiplier subsequence in $\mathcal{D} \times \mathbf{R}^m$ give KKT. Local inner minima alone may remain infeasible.
- For ALM, KKT + LICQ + SOSC gives local exactness at the correct multiplier for sufficiently large finite ρ; local convergence uses the nearby minimizing branch.

Extend to inequalities. For $\mathbf{h} \leq \mathbf{0}$, use the following added terms and multipliers:

$$\text{QP}: \quad \tfrac{\rho}{2}\|[\mathbf{h}]_+\|^2, \qquad \widehat{\boldsymbol{\nu}} = \rho[\mathbf{h}]_+,$$
$$\text{ALM}: \quad \sum_j \frac{[\nu_j + \rho h_j]_+^2 - \nu_j^2}{2\rho}, \quad \nu_j^+ = [\nu_j + \rho h_j]_+.$$

Also test dual feasibility and complementarity.

1. **A convex KKT certificate with an active zero multiplier.** Consider

$$\min_{\mathbf{x}\in\mathbf{R}^3} \frac{1}{2}\|\mathbf{x}-\mathbf{c}\|^2 \quad \text{s.t.} \quad \mathbf{e}^T\mathbf{x} = 4, \qquad \mathbf{x} \geq \mathbf{0},$$

where $\mathbf{c} = (4, 2, 1)^T$ and $\mathbf{e} = (1, 1, 1)^T$.

   (a) Show that a unique global minimizer exists and that LICQ holds at every feasible point.

   (b) Write and solve the KKT conditions, using $h_i(\mathbf{x}) = -x_i$. Find all multipliers and the active set. Does strict complementarity hold?

2. **Hard and soft margins.** The observations are

$$\mathbf{x}_1 = (-2, 0)^T,\ y_1 = -1; \qquad \mathbf{x}_2 = (1, 0)^T,\ y_2 = 1; \qquad \mathbf{x}_3 = (3, 1)^T,\ y_3 = 1.$$

   (a) Write the hard-margin SVM and its complete KKT conditions.

   (b) Use the first two constraints to obtain a lower bound on w_1. Find an optimal classifier and a multiplier certificate, and report its separator, margin width and loaded support vectors.

   (c) Add a fourth observation $\mathbf{x}_4 = (1, 0)^T$, $y_4 = -1$. Explain why hard margin is now infeasible. Write the soft-margin model with $C > 0$ and give one feasible point; you need not solve this model.

3. **A QP path and its cost.** Consider the equality-constrained problem

$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = x_1^2 + x_2^2 \quad \text{s.t.} \quad g(\mathbf{x}) = x_1 + x_2 + 1 = 0.$$

   (a) Find the constrained minimizer and its multiplier. For $P_\rho = f + \rho g^2/2$ ($\rho \geq 0$), find the exact inner minimizer $\mathbf{x}(\rho)$, the estimate $\widehat{\lambda}(\rho) = \rho g(\mathbf{x}(\rho))$, and their limits as $\rho \to \infty$.

   (b) Find the eigenvalues and condition number of $\nabla^2 P_\rho$. At $\mathbf{x}(\rho)$ compute $r_{\text{stat}}, r_{\text{feas}}, r_{\text{KKT}}$ using the estimated multiplier. How large must ρ be to attain $\varepsilon_{\text{out}} = 10^{-3}$? Explain why exact inner stationarity alone is an incomplete stopping test.

4. **Fixed-penalty ALM.** Use the problem in Question 3 and $L_A(\mathbf{x}, \lambda; \rho) = f(\mathbf{x}) + \lambda g(\mathbf{x}) + \rho g(\mathbf{x})^2/2$. Throughout this question $\rho > 0$ is fixed.

   (a) With exact inner minimization, derive the returned point $\mathbf{x}^{(k+1)}$ and the multiplier recurrence. Show that the multiplier error contracts and that $\{\mathbf{x}^{(k)}\}$ converges to the constrained solution.

   (b) Set $\rho = 1$, $\lambda^{(0)} = 0$. Compute the first two returned points and updated multipliers. Compare feasibility and inner Hessian conditioning with Question 3.